ACTU 301
Life Contingencies 1

Multiple State Models

Lecture 7

Introduction

What are multi-state transition models?

Probability models that describe the random movements of a subject among various states.

- We now introduce multiple state models of life contingent risks.
- Provide several examples of applications of "states" in actuarial contexts
- Describe the probabilistic mechanism for transiting from one state to another, in both discrete and continuous time, and
- Summarize important quantities that can be used for risk management such as the expected present value of insurance benefits and annuities, as well as policy values.
- Show how two life mortality can be viewed as a special case of the multiple state framework.

1. Examples of Multiple State Models

Consider a life aged x who, at time point t (age $x+t$) is in one of $n+1$ (*multiple*) states. We label the potential outcomes, known as the *state space*, as $0,1,\ldots,n$. Specifically, we let $Y(t)$ (sometimes denoted as $Y(x+t)$) be a categorical random variable with potential outcomes $0,1,\ldots,n$

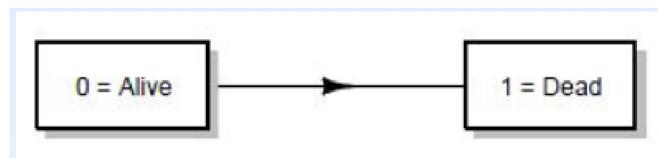
Special Case 1. The Alive-Dead Model.

In this special case, $Y(t)$ can take on values

- 0, meaning alive
- 1, meaning dead.

That is, there are two states under consideration, 0 and $n=1$. A graphical representation is given in Figure 1 where the arrow indicates that it is possible to transit from state 0 to state 1 (but not the reverse).

Potential Transitions in the Alive-Dead Model



Special Case 2. The Accidental Death Model.

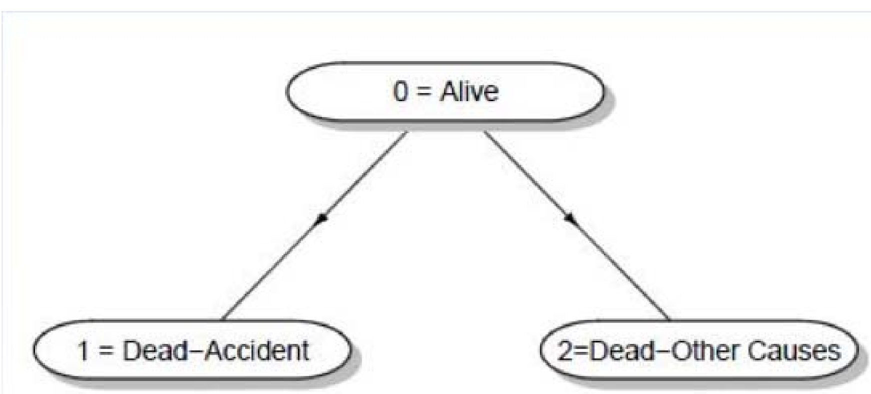
In this special case, $Y(t)$ can take on values

- 0, meaning alive
- 1, meaning death due to accident
- 2, meaning death due to all other causes.

This model is relevant for certain life insurance products that provide an extra benefit in the event of accidental death. There are three states under consideration, 0 , 1 and n =2.

A graphical representation is given below, where the arrows indicates that it is possible to transit from state 0 to either state 1 or state 2. It is not possible to transit out of state 1 or 2. If it is not possible to leave a state, it is referred to as an *absorbing state*.

Potential Transitions in the Accidental Death Model



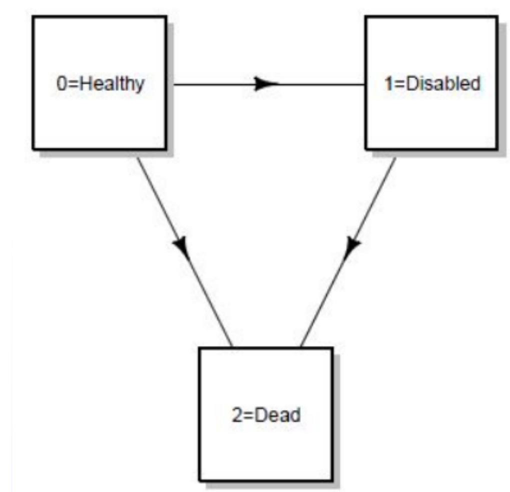
Special Case 3. Permanent Disability Model.

In this special case, $Y(t)$ can take on values

- 0, meaning health
- 1, meaning permanently disabled
- 2, meaning death.

This model is relevant for certain life insurance products that provide an extra benefit in the sickness such as a permanent disability. For example, it is common for some life insurance products to ``forgive'' premium payments during this period. In this model, disability is viewed as ``permanent'' in the sense that once a person achieves the disability status, recovery to the former ``health'' status is not possible.

Potential Transitions in the Permanent Disability Model

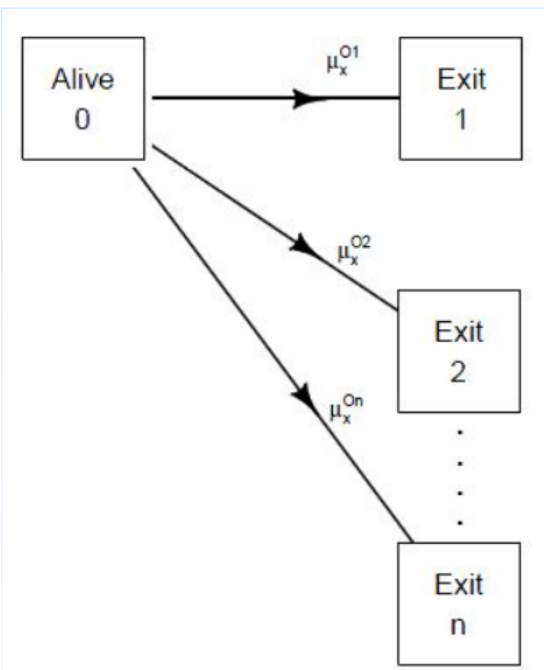


Special Case 4. Multiple Decrement Model.

In this special case, Y(t) can take on values 0,1,2,…,n where each of 1,2,…,n are absorbing states. The Alive-Dead and the Accidental Death Model are both special cases of the Multiple Decrement Model.

Because of its importance in applications, we examine multiple decrement models in a separate module.

Potential Transitions in the Multiple Decrement Model



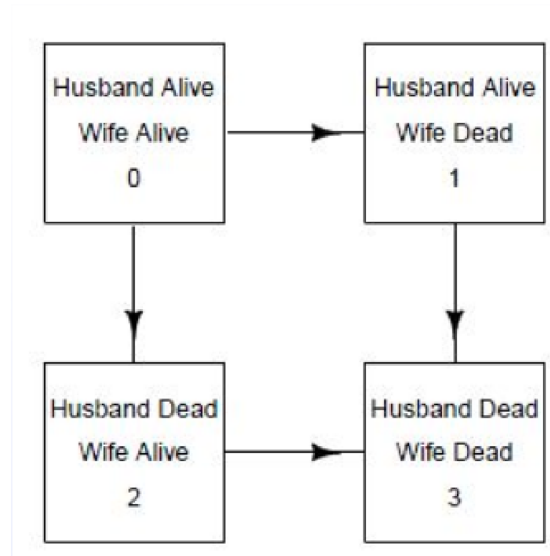
Special Case 5. Joint Life Model.

In this special case, we consider not one but two lives, e.g., *x* and *y*. For many applications, these represent ages at contract issue of a husband and wife for life and insurance or annuity products. In this special case, *Y*(*t*) can take on values

- ✓ 0, meaning both *x* and *y* are alive
- ✓ 1, meaning *x* is alive although *y* is dead
- ✓ 2, meaning *y* is alive although *x* is dead
- ✓ 3, meaning both *x* and *y* are dead.

Because of its importance in applications, we examine joint life models in a separate module.

Potential Transitions in the Joint Life Model of Two Lives



2. Multiple State Concepts

Our objective is to follow the categorical random variable $Y(t)$ over time t

We start by reviewing a few terms.

- A *stochastic process* is a collection of random variables.
- A *continuous-time stochastic process* is a stochastic process that is indexed by a continuous time variable such as time.
  - For example, we will use $\{Y(t)\}_{t\geq 0}$ to denote the history of states that (x) is in.
  - This is sometimes known as *continuous-time, discrete-state stochastic process* because the time index t is continuous but the outcome at any point in time, $Y(t)$ is discrete.
- A *discrete-time stochastic process* is a stochastic process that is indexed by a discrete time variable.
  - For example, we will use $\{Y(k)\}_{k=0,1,2,\ldots}$ to track the state of a person at regular intervals, e.g. monthly or annually
- For contrast, many viewers are familiar with (geometric) Brownian Motion $\{S(t)\}_{t\geq 0}$, a *continuous-time, continuous-state stochastic process*. Here, the time index t is also continuous. Further, $S(t)$, that can be interpreted as the stock price at a moment in time (\`\`S'' for stock price), is modeled as a random variable with *continuous* outcomes.

3. Probability Fundamentals

We begin with some basic definitions associated with probabilities of transitions. Then, we specialize to the (a) discrete time and (b) continuous time frameworks.

Probability Definitions

Transition Probabilities

Let $Y(t)$ be a discrete random variable. We are concerned with probabilities of the form

$$\Pr(Y(8) = 2|Y(4) = 0).$$

This is the probability that Y is in state 2 at time t=8, given that Y was in state 0 at time t=4. We will think of think as *transition probability*, that is, the probability of moving to a potentially new state, given some initial information.

More generally, consider the following transition probability

$$\Pr(Y(t_2) = i_2|Y(t_1) = i_1).$$

Here, we typically work with $t_1<t_2$. The states i_1 and i_2 both belong to the set of potential states, $\{0,1,\ldots,n\}$.

We might wish to calibrate the following probability

$$\Pr(Y(t_4) = i_4|Y(t_1) = i_1, Y(t_2) = i_2, Y(t_3) = i_3).$$

What is the probability that $Y(t_4)=i_4$ given the events $\{Y(t_1) = i_1\}$, $\{Y(t_2) = i_2\}$, and $\{Y(t_3) = i_3\}$? We use the following assumption to answer this complex question.

Markov Property

Markov Property (in words). Suppose you are given that the process is currently in state *i* at time *t* (e.g., $Y(t)=i$), where state *i* and time *t* are arbitrary. Then, the (conditional) probability of the process being in any state in the future does not depend on the past; it depends only on the current state.

Markov Property (using notation). Consider an arbitrary set of n+1 time points $t_1<\cdots<t_n<t$ and corresponding set of states $i_1,\ldots,i_n$, i such that

$\Pr(Y(t_1) = i_1,\ldots,Y(t_n) = i_n,\ Y(t) = i) > 0$.

Then, for future time length $s > 0$ and state j,

$\Pr(Y(t+s) = j \,|{=}Y(t) = i,\ Y(t_1) = i_1,\ldots,Y(t_n) = i_n) = \Pr(Y(t+s) = j \mid Y(t)=i)$.

Example: Alive-Dead Model

In this case n=1, i=0 for alive and j =1 for dead. Then we have

$${}_tp_x^{01}=\Pr(Y(x+t) = 1 \mid Y(x) = 0) = {}_tq_x,$$

our familiar probability of death for (x) with t years.

Probability Notation.

Let ${}_tp_x^{ij}$ be the probability that a life aged x in state i is in state j at age x+t. That is,

$${}_tp_x^{ij} = \Pr\left(Y(x+t) = j | Y(x) = i\right)$$

Probabilities, of course, are bounded by 0 and 1. Moreover, we consider only n+1 potential outcomes, so that

$${}_tp_x^{i0} + {}_tp_x^{i1} + \cdots + {}_tp_x^{in} = 1$$

For annuity calculations, it is helpful to define the notion that a life remains in a state over an extended period of time. Thus, we define

$${}_tp_x^{\overline{\overline{ii}}} = \Pr(Y(x+s) = i, \qquad for\ all\ s\ in\ [0,t) | Y(x) = 1)$$

that is, the probability that a life aged x *stays* in state i throughout the period from age x to age x+t. Note that ${}_tp_x^{\overline{\overline{ii}}} \leq {}_tp_x^{ij}$

. This is sometimes known as an *occupancy probability*.

Notation and Formulae

Notation

${}_tp_x^{ij}$ Probability of going from state *i* at age x to state *j* at age x + t

${}_tp_x^{ii}$ Probability of remaining in state *i* for a period *t* for an individual aged *x* .

${}_t\mu_x^{ij}$ Rate of changing from state i to state j for an individual aged x in state i (i $\neq$ j).

Formulae

$${}_tp_x^{ij} = (Y(x+t) = j | Y(x) = i)$$

$${}_tp_x^{ii} = P(Y(x+s) = i \text{ for all } 0 \le s \le t | Y(x) = i)$$

$$\mu_x^{ij} = \lim_{t \to 0^+} \frac{{}_tp_x^{ij}}{t}$$

Formulae for Probabilities

$${}_tp_x^{\overline{ii}} = e^{-\int_x^{x+t} \sum_{j \neq i} \mu_y^{ij} dy}$$

$${}_{t+s}p_x^{ij} = \sum_k {}_tp_x^{ik} \, {}_sp_{x+t}^{kj}$$

$$\frac{d}{dt} {}_tp_x^{ij} = \sum_{k \neq j} {}_tp_x^{ik} \mu_x^{kj} - {}_tp_x^{ij} \mu_x^{jk}$$

Discrete Time Fundamentals

To develop intuition, in this section we consider both ages x and duration time t as discrete. When probabilities do not depend on the conditioning time (age), Markov probabilities are said to be (time-) *homogenous*.

This special case is helpful in particular for short-term coverages such as some types of health insurance as well as property and casualty insurance. When probabilities change with time (age) (the usual case for mortality), Markov probabilities are said to be (time-) *inhomogenous*.

Many calculations are simply complicated applications of the familiar *Law of Total Probability*. To see how this works, consider the following.

Example: Rain

Suppose that there only two states of the world,

0 = *Rain* (*R*) and
1 = *No Rain* (*NR*).

You are given that

- ✓ if it is raining today, then the probability of rain tomorrow is 0.7
- ✓ if it is not raining today, then the probability of no rain tomorrow is 0.6.

Assume that is raining today at time t=1, what is the probability that it will be raining at time t=3?

Solution:

By the law of total probability,

$$\Pr[Y(3)=R|Y(1)=R] = \Pr[Y(2) = R|Y(1) = R] \times \Pr[Y(3) = R|Y(2) = R] + \Pr[Y(2) = NR|Y(1) = R] \times \Pr[Y(3) = R|Y(2) = NR]$$

$$= (0.7)\,(0.7) + (0.3)\,(0.4) = 0.61.$$

Chapman-Kolmogorov Equation

In notation, we can express this as

$$\Pr[Y(3)=0|Y(1)=0] = \Pr[Y(2)=0|Y(1)=0] \times \Pr[Y(3)=0|Y(2)=0] + \Pr[Y(2)=1|Y(1)=0] \times \Pr[Y(3)=0|Y(2)=1]$$

$${}_2p_1^{00} = {}_1p_1^{00} \text{ x } {}_1p_2^{00} + {}_1p_1^{01} \text{x } {}_1p_2^{10}$$

or simply as

$${}_2p_1^{00} = p_1^{00} \text{ x } p_2^{00} + p_1^{01} \text{x } p_2^{10}$$

where we have dropped the "1" prefix, as is common in actuarial notation.

Example - Rain - Continued

Assuming that is not raining today at time t=1, what is the probability that it will be raining on day t=3?

Solution

$$ {}_2p_1^{10} = \ p_1^{10} \text{ x } \ p_2^{00} + \ p_1^{11} \text{x } \ p_2^{10} $$

= (0.4)(0.7) + (0.6)(0.4) = 0.52

Chapman-Kolmogorov equation

More generally, we have

$$ {}_2p_1^{ij} = \sum_{k=0}^{n} p_1^{ik} \text{ x } p_2^{kj} $$

Next, convince yourself that

$$ {}_3p_1^{01} = \sum_{k=0}^{n} p_1^{0k} \text{ x } \ {}_2p_2^{k1} $$

and

$$ {}_3p_1^{01} = \sum_{k=0}^{n} {}_2p_1^{0k} \text{ x } \ p_3^{k1} $$

and

$$ {}_5p_1^{ij} = \sum_{k=0}^{n} {}_2p_1^{ik} \text{ x } \ {}_3p_3^{kj} $$

These are all special cases of the **Chapman-Kolmogorov equation**

$$ {}_{m+r}p_x^{ij} = \sum_{k=0}^{n} {}_mp_x^{ik} \text{ x } \ {}_rp_{x+m}^{kj} $$

Example

Using the Rain Example probabilities, calculate the probability that if it is raining on Monday (at t =1) then it will not be raining on Friday (at t = 5).

Solution

Begin with the Chapman-Kolmogorov equation, choose $m + r = 4$, $i = 0$ and j =1, to get

$$ {}_4p_1^{01} = \sum_{k=0}^{n} {}_mp_1^{0k} \text{ x } {}_rp_{m+1}^{k0} $$

There are different choices for m and r. We use $m = r = 2$, to get

$$ {}_4p_1^{01} = \sum_{k=0}^{n} {}_2p_1^{0k} \text{ x } {}_2p_3^{k0} $$

We determine

${}_2p_1^{00} = p_1^{00} \text{ x } p_2^{00} + p_1^{01} \text{x } p_2^{10}$ = (0.7)(0.7)+(0.3)(0.4)=0.61

$$ {}_2p_1^{01} = 1 - {}_2p_1^{00} = 0.39 $$

${}_2p_1^{10} = p_1^{10} \text{ x } p_2^{00} + p_1^{11} \text{x } p_2^{10}$=(0.4)(0.7)+(0.6)(0.4)=0.52

$$ {}_2p_1^{11} = 1 - {}_2p_1^{10} = 0.48 $$

Thus,

$$ {}_4p_1^{01} = {}_2p_1^{00} \text{ x } {}_2p_3^{01} + {}_2p_1^{01} \text{x } {}_2p_3^{11} $$

=(0.61)(0.39) + (0.39)(0.48) = 0.4251

Example

The Simple Insurance Company starts at time 0 with a surplus of 3. At the beginning of every year, it collects a premium of 2. every year, it pays a random claim amount as shown:

Claim Amount	0	1	2	4
Probability of Claim Amount	0.15	0.25	0.40	0.20

If, at the end of the year, Simple's surplus is more than 3, it pays a dividend equal to the amount of surplus in excess of 3. If Simple is unable to pay its claims, or if its surplus drops to 0, then it goes out of business. Simple has no administrative expenses and interest is equal to 0.

Calculate the probability that Simple will be in business at the end of three years.

Solution

We follow surplus for the state space that may be in one of 0,1,2, or 3. Note that Surplus = 0 is an *absorbing* state, there is no probability of exit so that ${}_tp_m^{00} = 1 \; and \; {}_tp_m^{0j} = 0$, for j =1,2,3.

We start at time 0 with a surplus of 3 and so, the probability of surplus being zero at the end of three years can be denoted as ${}_3p_0^{03}$. With this notation, the probability of being in business at the end of three years is $1 - {}_3p_0^{03}$.

Now, with the Chapman-Kolmogorov equation, choose m =1,r =2, l =3 and j = 0, to get

$$ {}_3p_1^{30} = \sum_{k=0}^{3} p_0^{3k} \text{ x } {}_2p_1^{k0} $$

To plug numbers into the Chapman-Kolmogorov equation, it is helpful to write down the set of transition probabilities:

Surplus Transition Probabilities				
"From" or	"To" or			
"Origin State *i*"	"Destination State *j*"			
Surplus	0	1	2	3
0	1	0	0	0
1	0.2	0.4	0.25	0.15
2	0.2	0	0.4	0.4
3	0	0.2	0	0.8

Where do these numbers come from? Well, start with the last row, corresponding to i=3. At the start of the year, we have 3 of surplus plus 2 more in premiums, for a total of 5. If we get 4 in claims, then we drop down to 1 with a probability of 0.2, or p_1^{31}=0.2. If we get 0,1, or 2 claims, then the surplus before dividend is 5, 4 and 3, respectively, so the surplus after dividends is 3.
Thus, p_1^{33}=0.15+0.25+0.40=0.8. This takes care of all the events in that row that can happen, so the other probabilities are 0. Use similar logic to confirm the probabilities in the other rows.

Now, going back to the Chapman-Kolmogorov equation, we see that

$$ {}_3p_1^{30} = p_0^{30} \text{ x } {}_2p_1^{00} + p_0^{31}\text{x } {}_2p_1^{10} + p_0^{32} \text{ x } {}_2p_1^{20} + p_0^{33}\text{x } {}_2p_1^{30} $$
$$ = (0.2) \text{ x } {}_2p_1^{10} + (0.3)\text{x } {}_2p_1^{30} $$

and

$$ {}_2p_1^{10} = p_0^{10} \text{ x } p_1^{00} + p_0^{11}\text{x } p_1^{10} + p_0^{12} \text{ x } p_1^{20} + p_0^{13}\text{x } p_1^{30} $$
$$ = (0.2)(1) + (0.4)(0.2) + (0.25)(0.2) + (0.15)(0) = 0.33 $$

$$ {}_2p_1^{30} = p_0^{30} \text{ x } p_1^{00} + p_0^{31}\text{x } p_1^{10} + p_0^{32} \text{ x } p_1^{20} + p_0^{33}\text{x } p_1^{30} $$
$$ = (0) + (0.2)(0.2) + 0 + (0.8)(0) = 0.04 $$

Thus

$$ {}_3p_1^{30} = (0.2)\text{x } {}_2p_1^{10} + (0.8) \text{ x } {}_2p_1^{30} $$

$$ = (0.2)(0.33) + (0.8)(0.4) = 0.098 $$

This yields the probability of being in business at the end of three years is

$$ 1 - {}_3p_1^{30} = 1 - 0.098 = 0.902 $$

Continuous Time Fundamentals

Now return to the more general case where we think about both age *x* and time *t* as evolving on a continuous basis.

For $i \neq j$, define

$$\mu_x^{ij} = \lim_{h \to 0^+} \frac{{}_h p_x^{ij}}{h}$$

to be the *force of transition* from state *i* to state *j*. It is also known as the *transition intensity*.

Special Case: Alive-Dead Model

In this case $n = 1$, $i = 0$ for alive, $j = 1$ for dead, and ${}_h p_x^{01} = {}_h q_x$. Thus,

$$\mu_x^{01} = \lim_{h \to 0^+} \frac{{}_h p_x^{01}}{h} = \lim_{h \to 0^+} \frac{{}_h q_x}{h}$$

our usual definition of the force of mortality.

"Little o" Notation. Some authors prefer to use notation to describe infinitesimal terms. When we use the notation "*o*(1)", this stands for a sequence of terms that converges to 0 along some index. For example, think of the notation "*o*(1)" as meaning a sequence of terms, $\{\epsilon_h\}$, where

$$\lim_{h \to 0} \epsilon_h = 0$$

In the same way, "*o*(*h*)" stands for a sequence of terms that converges to 0 faster than an index *h*, that is, $\lim_{h \to 0} \frac{\epsilon_h}{h} = 0$. More generally, "*o*(*g*(*h*))" stands for a sequence of terms, $\{\epsilon_h\}$, where $\lim_{h \to 0} \frac{\epsilon_h}{g(h)} = 0$

How do we evaluate transition probabilities based on transition intensities? We will introduce two results, one for occupancy probabilities and one known as *Kolmogorov's forward equation*.

Occupancy Probabilities

With the transition intensities, we can compute occupancy probabilities with:

$$ {}_tp_x^{\overline{\overline{ii}}} = exp\left\{-\int_0^t \sum_{j=0,j\neq i}^{n} \mu_{x+s}^{ij} ds\right\} $$

Special Case: Alive-Dead Model

In this case n=1, i =0 for alive, j =1 for dead, and ${}_hp_x^{01}$= ${}_hq_x$. Thus,

$$ \mu_x^{01} = \lim_{h\to 0^+} \frac{{}_hp_x^{01}}{h} = \lim_{h\to 0^+} \frac{{}_hq_x}{h} = \mu_x $$

our usual definition of the force of mortality. Thus,

$$ {}_tp_x = {}_tp_x^{\overline{00}} = exp\left\{-\int_0^t \sum_{j=1}^{1} \mu_{x+s}^{0j} ds\right\} = exp\left\{-\int_0^t \mu_{x+s}^{01} ds\right\} $$

Kolomogorov's Forward Equation

A second method to evaluate probabilities is based on *Kolmogorov's forward equation*:

$$ \frac{\partial}{\partial t}\ {}_tp_x^{ij} = \sum_{k=0,k\neq j}^{n} \left({}_tp_x^{ik}\mu_{x+t}^{kj} - {}_tp_x^{ij}\mu_{x+t}^{jk} \right) $$

Special Case: Multi-Decrement Model

In this case, the positive probabilities only occur at the origin state i=0. Similarly, forces of transition are only positive when the origin state is 0. Thus,

$$ \frac{\partial}{\partial t}\ {}_tp_x^{0j} = {}_tp_x^{00}\mu_{x+t}^{0j} $$

Integrating both sides yields

$$ {}_tp_x^{0j} = \int_0^t {}_sp_x^{00}\mu_{x+s}^{0j} ds $$

4. Insurance Benefits

For insurances, suppose that a life age x is in state i at time 0 and that 1 is paid immediately at time t for transition to state k. Then, the EPV is

$$\bar{A}_x^{ik} = \int_0^\infty \sum_{j \neq k} e^{-\delta t} \, {}_tp_x^{ij} \mu_{x+t}^{jk} \, dt$$

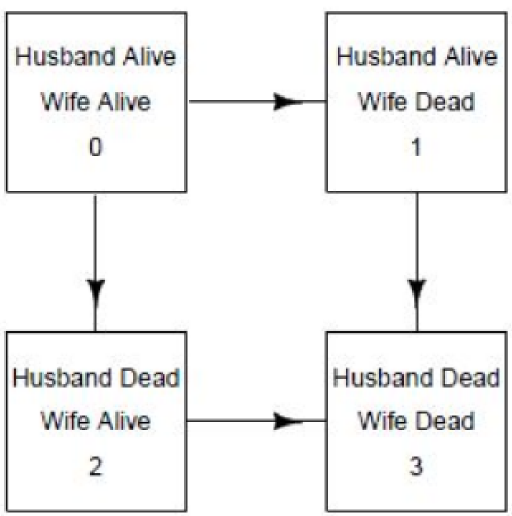
This assumes that the person transits from any state j other than k.

5. Special Case: Two Lives

Joint and last survivor probabilities can be expressed as a special case of the multiple state formulation and so no new theory is required to handle this special case. Define the following four states:

- 0 means both x and y are alive
- 1 means x is alive although y is dead
- 2 means y is alive although x is dead
- 3 means both x and y are dead.

Joint Life Status for Two Lives



Traditional Joint Life Notation

Notation		
Traditional	**Multi-State**	**Concept**
${}_tp_{xy}$	${}_tp^{00}_{xy}$	Pr[*x* and *y* are both alive in *t* years]
${}_tq_{xy}$	${}_tp^{01}_{xy} + {}_tp^{02}_{xy} + {}_tp^{03}_{xy}$	Pr[*x* and *y* are not both alive in *t* years]
${}_tq^{1}_{xy}$		Pr[*x* dies before *y* in *t* years]
${}_tq^{2}_{xy}$		Pr[*x* dies after *y* in *t* years]
${}_tp_{\overline{xy}}$	${}_tp^{00}_{xy} + {}_tp^{01}_{xy} + {}_tp^{02}_{xy}$	Pr[at least one of *x* and *y* are alive in *t* years]
${}_tq_{\overline{xy}}$	${}_tp^{03}_{xy}$	Pr[*x* and *y* are both dead in *t* years]

In the traditional notation, the right subscript refers to the status, e.g. xy. The p-type probabilities refer to the survival of the status - the q-type probabilities refer to the failure of the status. The numbers on top of ages indicate that the status is contingent, where order of failure is important. Although there is no immediate expression, it can be calculated using multiple state concepts as:

$$ {}_tq^{1}_{xy} = \int_0^t {}_sp^{00}_{xy}\mu^{02}_{x+s:y+s}ds = \int_0^t {}_sp_{xy}\mu_{x+s}ds $$

similarly

$$ {}_tq^{2}_{xy} = \int_0^t {}_sp^{01}_{xy}\mu^{13}_{x+s}ds = \int_0^t {}_sq_y\ {}_sp_x\mu_{x+s}ds $$

END